

LUÍS M. C. D. CENTENO

Planetary and Astrobiology Researcher

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PROFILE

Geoscientist & Planetary Researcher. Recent Master of Space Studies graduate (ISU) specializing in **planetary surface/subsurface geology** and **autonomous spectroscopic analysis**. I combine an academic background in **Earth sciences** with hands-on expertise in **analogue fieldwork**, including cave exploration and Arctic expeditions. Developing **autonomous machine learning pipelines for planetary surface rovers** (ExoMars Rosalind Franklin) and **integrating AI with mineralogical datasets for autonomous robotic decision-making**. Highly interested in advancing ESA's PANGAEA analytical toolsets to enable reliable, autonomous biosignature detection in Martian subsurface environments.

EDUCATION

International Space University | Sep-25 to Jun-26 | Strasbourg, FR

Master of Space Studies - Space Science & Exploration

Key Studies: Planetary Exploration, Lunar Geology, Martian Hydrology, AI-Spectroscopy Integration, Planetary Seismology, and Engineering Subsystems.

- Lead Scientist | ISU M4 Team Project: Developed a **data integration framework** for **active seismic sensing** and **autonomous mineral mapping** on Mars.
- AI Integration: Engineered **ML simulations for mineralogical data analysis** to optimize **autonomous robotic decision-making** for volatile detection.
- Planetary Research: Quantifying **Martian Hesperian hydrological evolution** by integrating alluvial geomorphology with quantitative crater-fill data.
- Astrophysics & Data Mining: Established a framework for early universe ($z=14$) evolutionary models using **JWST/ALMA spectral data**; conducted large-scale **stellar data mining analysis** to produce CMDs for the Cluster Cl Pismis.
- Geospatial Data: Developed **JMARS** workflows to **classify** and **determine volatile-rich sediments** in crater deposits for **optimal landing site selection for future Mars missions**.

UNIVERSITY OF STAVANGER | Aug-16 to Jun-18 | Stavanger, NO

MSc Petroleum Geosciences Engineering - Specialization: Subsurface Modeling

Key Studies: Advanced Seismic 2D/3D Modeling, Geophysics, and Geochemistry.

- Subsurface Data Interpretation: Expert in building **2D & 3D structural models** and **interpreting seismic reflection data** using industry-standard tools (e.g., PETREL, MOVE).
- Quantitative Modeling: Trained in **lab sample geochemical handling**, petrophysical analysis, and stratigraphic modeling to quantify subsurface volumes.
- Achievement: Led the Norwegian national **winning team** for the 2017 AAPG Imperial Barrel Award in **processing raw datasets** of subsurface characterization and complex basin analysis.

BIRKBECK UNIVERSITY | Sep-21 to Jun-23 | London, UK

Graduate Certificate in Planetary Sciences

Key Studies: **Astrobiology, Planetary Spectroscopy**, Astronomy, Geology of the Solar System.

FACULTY OF SCIENCE, UNIVERSITY OF LISBON | Sep-10 to Jun-16 | Lisbon, PT

BSc Geology & Post-Graduation in Subsurface Geosciences

Key Studies: **Geology**, Tectonophysics, **Mineralogy**, **Sedimentology**, Reservoir modeling, Advanced Fieldwork

2023 -2024 | London, UK & Lisbon, PT

Complementary Education

- IE BUSINESS SCHOOL **Data Science - Data science & analysis**, machine learning, and statistical modeling (2 months).
- INGEOEXPERT **Advanced Subsurface Analysis** - Seismic Interpretation (2D/3D), Subsurface Data Analysis; Thermo-chronology and Tectonic Modeling (3 months).

WORK EXPERIENCE

Industry: Geospatial, Data & Project Management, AI

LUNEX EuroMoonMars (ILEWG) | July to Dec 26 | Wassenaar, NL

Astrobiology & Planetary Research Intern

- **Planetary Spectrometry & Data Analysis:** Analysing orbital datasets from the Mars Express High Resolution Stereo Camera (HRSC) alongside OceanOptics USB4000 spectrometer datasets to **characterize habitability in targeted Martian landing sites**.
- **Analogue Mission Operations:** Conducting analogue astronaut fieldwork in Mars-like environments, **testing spectrometers for the in-situ astrobiological and mineralogical analysis** of terrestrial geological analogues.
- **Instrument & Mission Support:** Contributing to **scientific target identification research** for the ExoMars Panoramic Camera (PanCam) with **Dr. Bernard Foing** to optimize autonomous robotic operations, while assisting in early-stage lunar orbit spectrometer mission planning.
- **Scientific Dissemination & AI Integration:** First-authored three **research papers accepted** at major international forums (**EANA, EPSC, IAC**), focusing on autonomous AI mineral classification for ExoMars navigation, subsurface hydrology, and active seismic sensing.
- **Strategic Liaison:** Facilitated collaborative frameworks between EuroMoonMars and prospective aerospace institutes (**including ESA**), acting as the primary institutional bridge for future research tracks.

GEOGLOBAL | Apr-19 to Sep-23 | Lisbon, PT

Geologist & Project Coordinator (Geospatial Intelligence)

- Coordinated the **€2.9M COSMO project**, managing a **multidisciplinary team** to translate **geospatial datasets** into strategic environmental frameworks, resulting in positive **environmental policy changes**.
- **Data Fusion: Merged bathymetric** (acoustic) **surveys** with high-resolution **aerial imagery** to **model sedimentation patterns** and regional **geomorphological evolution** over a 3-year lifecycle.
- Architected **spatial data pipelines** and **Digital Terrain Models (DTMs)**, ensuring **data integrity** and precision for **large-scale geospatial analysis** and planning.

POLT, AI Information services | Nov-23 to June-25 | Lisbon, PT

Technical Director & AI Systems Lead

- Led a multidisciplinary team to **develop** in-house **Machine Learning software** for **large-scale statistical database analysis**, focusing on "maximum truth-seeking" information. Managed technical workflows and **interdisciplinary communication**.

Ciência Viva & Pt. Speleology Society (SPE) | Aug-13 to Sep-16 | Lisbon, PT

Science Operations & Subsurface Exploration

- Subsurface Exploration & Mapping: **Conducted extensive geological mapping** and surveys in complex **cave systems**, identifying new paths and skylights through **fieldwork missions**.
 - Scientific Sample Collection: Executed systematic subsurface **sample collection for geochemical analysis**, ensuring data integrity for **academic research** and **planetary analog studies**.
 - Laboratory & Academic Support: Served as a **Lab Assistant in Geochemistry** and a **Structural Geology Assistant Teacher**, managing experimental protocols and teaching advanced fieldwork methodology.
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PUBLICATIONS & CONFERENCES

- **In Preparation:** Centeno, L. & Sounny, A. "Comparative Study of Martian and Terrestrial Megafans to Understand Past Hydrology."
 - Directly supports the expansion and curation of planetary reference libraries for sedimentary analog structures.
- **European Astrobiology Network Association (EANA) | First Author:** "Targeting Martian Biosignatures: Autonomous AI Mineral Classification and Rover Navigation for ExoMars."
- **Europlanet Science Congress (EPSC) | First Author:** "Impact Craters as Hydrologic Refugia: Quantifying Subsurface Water-Ice Volumes in Martian Alluvial-Fan Hosting Basins using Terrestrial Analogues."
- **International Astronautical Congress (IAC) | First Author:** "A Rover-Drone Approach to Martian Lava Tubes Exploration: Integrating Active Seismic Sensing and Optical Aerial Surveying."
- **Geosciences Journal (2019):** "Impact of Salt Tectonics on Thermal Evolution and Petroleum Systems in confined Rift Basins."

SKILLS

Geophysics & Subsurface Modeling	2D/3D Seismic Interpretation, Seismic Wave Theory, Seismic Profiling, Tectonic & Stratigraphic Modeling, and Petrophysical Analysis.
Data Science & IT Tools	JMARS, Google Earth Engine, STK, Digital Terrain Modeling (DTM), UAV Data Collection, ArcGIS/QGIS, Python, MATLAB, Machine Learning, and AutoCAD
Planetary Science & Exploration	Lunar Regolith Geomorphology, Martian Volatile Detection in "Cold Traps", Planetary Spectroscopy, Astrobiology, and Analog Fieldwork Methodology.
Data Science & AI	Machine Learning Development (Mineral & Feature Classification), Statistical Data Analysis, Signal Processing, and Automated Data Fusion.
Languages	Portuguese (Native), English (Fluent), Spanish (Fluent), French (Beginner)